

CARSON SILA WAMBUA

BACKEND DEVELOPER & OPERATIONAL DATA ANALYST | FINTECH MICROSERVICES & SRE INTELLIGENCE

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PROFESSIONAL SUMMARY

Versatile **Backend Software Engineer & Operational Data Analyst** with proven expertise in building high-throughput fintech microservices and mission-critical operational data systems. At **E&M; Technology House**, serves as core Backend Developer for the flagship **Distributor Financing Platform**, engineering end-to-end **Onboarding** and **Repayments** microservices utilizing Java 21, Spring Boot 3.4, Spring Cloud, M-Pesa/CBS payment rails, and waterfall allocation engines. Combines enterprise software engineering with Network Operations Center (NOC) experience at **Quavatel Limited** (99.8% SLA uptime for Safaricom, Airtel, Liquid Telecom) and end-to-end data analytics capability: creator of **ClimaCrop Intelligence** (Agri-Fintech de-risking platform — 40 crops, 35 counties, ML + rule-based suitability, DSCR underwriting) and **OpsPulse AI** (operational copilot cutting MTTR by 73%). Proficient in Java, Python, SQL, microservices architecture, ML, and agricultural analytics.

CORE TECHNICAL COMPETENCIES

Backend & Microservices:	Java 21, Spring Boot 3.4, Spring Cloud (Eureka, Config Server, Gateway), REST APIs, Feign Clients, Swagger/OpenAPI, Maven, Docker, Jenkins
Fintech & Payments:	Multi-Tenant Onboarding, KYC Document Processing, M-Pesa C2B/B2B APIs, Core Banking System (CBS) Callbacks, Waterfall Debt Repayment Strategies, Maker-Checker Governance
Data Analytics & ML:	Python (Pandas, NumPy, Scikit-Learn, SciPy), SQL (MySQL, PostgreSQL), Isolation Forest, K-Means Clustering, Random Forest, AEZ Rule-Based Engines, Streamlit, Plotly, Pytest
Agri-Tech & Climate:	TAHMO Weather Station Data (116 stations, 35 counties), Crop Suitability Modelling (40 crops), Agri-Fintech Underwriting (DSCR, Risk Index), KilimoBot AI (Gemini + offline), NASA POWER / FAOSTAT Ingest
Operations & Infrastructure:	NOC 24/7 Monitoring, SLA & MTTR Optimization, Root Cause Analysis (5-Whys), Fiber Transmission Restoration (FTTH/OTDR), Git/GitHub, Linux

PROFESSIONAL EXPERIENCE

Backend Software Developer — Distributor Financing Platform July 2026 – Present

E&M; Technology House — Financial Technology & Enterprise Engineering

Nairobi, Kenya

- Spearheaded backend architecture for the **Distributor Financing Platform**, a multi-tier closed-loop supply chain financing engine connecting commercial banks, anchor manufacturers, and FMCG distributors.
- Architected and developed the **Onboarding Microservice**: Implemented multi-tenant KYC workflows, document storage services, and dynamic RBAC security across Bank Admins, Manufacturers, and Distributors.
- Engineered the **Repayments Microservice**: Built high-reliability payment processing integrating **M-Pesa** and **Core Banking System (CBS)** debit callbacks with zero transactional discrepancy.
- Implemented dynamic **Waterfall Allocation Engine** (FeesInterestPrincipalStrategy) that prioritizes incoming liquidations across penalties, fees, interest, and principal, auto-recycling revolving credit headroom.
- Designed financial **Maker-Checker Approval Controls** and automated ledger reconciliation routines, maintaining **99.95% transaction integrity** across high-volume loan servicing events.
- Collaborated across cross-functional microservices (Loans, Notifications, Reports, and API Gateway), standardizing inter-service Feign communication and OpenAPI documentation.

Network Operations & Telecommunications Engineering Intern

April 2024 – October 2024

Quavatel Limited — Network Operations Center (NOC) & Field Operations

Nairobi / Thika, Kenya

- Monitored 24/7 network telemetry and fiber transmission backbone for tier-1 telecom operators (**Safaricom, Airtel, Liquid Telecom**), sustaining a **99.8% network SLA uptime** benchmark.
- Executed emergency optical fiber cable cut localization and precision fusion splicing under strict SLA countdown windows, accelerating average emergency restoration time by **28%**.
- Audited and maintained enterprise GPS fleet tracking and telematics software, verifying reporting uptime for 50+ field operational vehicles.
- Performed Fiber-To-The-Home (FTTH) network installations and optical power diagnostics, boosting first-visit resolution to **92%**.

Operational Data Analyst Fellow — Enterprise SRE & Analytics Program

June 2026 – Present

Power Learn Project (PLP) / Inuka Enterprise Analytics Track

Nairobi, Kenya

- Architected and deployed **OpsPulse AI**, an operational copilot processing 1,800+ monthly incident logs to predict SLA breaches 15 minutes in advance, reducing MTTR from 45 min to 12 min (**-73%**).
- Formulated CFO business case proving **461.4% Year 1 ROI** and **\$320,000** in annual savings across penalty avoidance and labor recovery with a **53-day payback period**.
- Implemented unsupervised machine learning (**Isolation Forest**) over multivariate incident telemetry to isolate operational anomalies with **94% precision**.

STRATEGIC FEATURED PROJECTS

Distributor Financing Backend Platform (E&M; Technology House)

Java 21, Spring Boot 3.4, Spring Cloud, MySQL

- Built the core **Onboarding** and **Repayments** microservices for closed-loop distributor financing, enabling banks to disburse invoice financing directly to manufacturers on behalf of distributors.
- Developed M-Pesa C2B/B2B and CBS payment callback listeners with automatic transaction idempotency, batch sweep schedulers, and multi-currency ledger reconciliation.

- Implemented Spring Security with JWT, Eureka service discovery, Spring Cloud Gateway with aggregated Swagger OpenAPI UI, and containerized Docker infrastructure.

Python, Streamlit, Scikit-Learn, Gemini AI, Docker

ClimaCrop Intelligence: Climate-Smart Agri-Fintech De-Risking Platform (Capstone)

- Built a role-based Streamlit platform serving **Cooperative Members, Bank/SACCO Officers, and Researchers** with personalized views of 10-year TAHMO climate data (116 stations, 35 counties) and a 40-crop agronomic suitability matrix.
- Engineered a dual-engine crop suitability pipeline: a transparent **Agro-Ecological Zone (AEZ) rule engine** (default) and a legacy Random Forest classifier (audit-only), each output carrying a **Provenance tag** (MEASURED/OFFICIAL/ESTIMATED/MODELED) so users know data confidence.
- Implemented **Agri-Fintech underwriting** engine: composite risk index $R_{agri} = 0.40(\text{Climate}) + 0.35(1-\text{CropFit}) + 0.25(\text{Market Volatility})$, automated loan facility sizing (70% CapEx), risk-adjusted interest (12% + 8% premium), and DSCR portfolio stress test.
- Integrated **KilimoBot AI assistant** (Google Gemini API + offline rule-based fallback) providing plain-language Swahili/English advisory and market arbitrage insights across 5 regional wholesale markets.

OpsPulse AI: Enterprise Operations Intelligence & SLA Copilot

Python, Streamlit, Scikit-Learn, Plotly

- Engineered an operational intelligence dashboard featuring real-time KPI benchmarking, multivariate **Isolation Forest** anomaly detection, and grounded SRE runbook retrieval (RAG), cutting MTTR by **73%** (45 min → 12 min).
- Automated 5-Whys post-mortems and CLI remediation checklists; maintained a 15-test Pytest suite at **100% pass rate**.

EDUCATION

Bachelor of Science in Information Technology

2024 – Nov 2026 (Expected)

Zetech University — Coursework Completed, Focus on Software Engineering & Data Analytics

Nairobi, Kenya

Diploma in Information Technology

2022 – 2024

Zetech University — Graduated with Distinction; Created Centralized Examination Management Portal

Nairobi, Kenya

HONORS, LEADERSHIP & VOLUNTEERING

- **1st Runner-Up (2nd Place Overall)**, University Innovation Hackathon (2025) — Co-led architecture for campus fintech solution.
- **Youth Ministry & Logistics Leader**, CITAM Thika (2021 – 2025) — Coordinated weekly operations and scheduling for 500+ attendees.
- **E-Commerce Agricultural Trade Platform** — Built degree capstone connecting smallholder farmers to markets, receiving top marks from university evaluation panel.